# Head-Gesture Game Jam

Event brief for organizers, volunteers and judges

**CONFIDENTIAL UNTIL 12:00 PM.** This brief reveals the twist (head-gesture control). Do not share it with participants before the reveal.

| | |
|---|---|
| **Date** | October 15, 2026 |
| **Time** | 8:00 AM to 2:00 PM (check-in from 8:00, event starts 9:00) |
| **Venue** | AB3 Gallery Classroom |
| **Organizer** | MIC Development Department, VIT Chennai |
| **Participants** | 25 teams of 4 members (100 participants) |
| **Prize** | 2 winning teams, chosen by 2 faculty judges. Each winning team gets one month of ChatGPT Plus. |

## Overview

Teams get **3.5 hours** to build a game controlled only by the **arrow keys, Z and Space**. Then they demo it using **head gestures instead of a keyboard**.

- Participants build the game, using AI tools and a starter prompt we provide.
- We build the controller: an OpenCV station that reads head tilt, mouth movement and blinks, and presses keys on the team's laptop over the network.
- **The twist:** teams learn about head control only in the last 30 minutes of the build, so they must adapt their game to imprecise, held inputs.

# Format

| Item | Detail |
|---|---|
| Teams | 25 teams × 4 members = 100 participants |
| Build window | 3.5 hours (9:00 AM to 12:30 PM) |
| Controls | Arrow keys, Z and Space only. Keys must work when **held**. |
| Tools | Any language or engine. AI coding tools allowed. |
| Starter prompt | Written by organizers. Teams fill in their theme, characters and plan in blank fields. |
| Twist reveal | Head-gesture control announced at 12:00 PM, the last 30 minutes of building |
| Demo | Up to 10 minutes per team at one of 5 stations |
| Players at demo | Team members and the judges |
| Winners | 2 teams, picked by 2 faculty judges. Each wins one month of ChatGPT Plus. |

# Timeline

Building runs 9:00 AM to 12:30 PM. All 25 demos fit in five 13-minute rounds before results at 1:50 PM.

| Time | What happens | Who |
|---|---|---|
| 8:00 - 9:00 | **Check-in.** Teams install the listener script and pass a connection test at the desk. | Desk volunteers |
| 9:00 | **Kickoff.** Rules explained, starter prompt shared, build begins. | Organizers |
| 9:00 - 12:00 | **Build.** Teams build with arrow keys, Z and Space. | Teams |
| 12:00 | **Twist reveal.** Head-gesture control announced and demonstrated on one station. | Organizers + 1 station |
| 12:00 - 12:30 | **Adapt and test.** Teams adjust their games. Volunteers run a quick key test on each laptop. | Teams + volunteers |
| 12:30 | **Build ends.** Code freeze. Stations calibrated. | Station operators |
| 12:40 - 1:45 | **Demos.** 5 rounds across 5 stations: 10 minutes of play plus 3 minutes of changeover each. | Teams, operators, judges |
| 1:50 | **Results.** Judges announce the 2 winning teams. | Judges |
| 2:00 | Event ends. | |

# Demo schedule

Five stations run in parallel. Each round is **13 minutes**: 3 minutes of changeover followed by 10 minutes of play. The station's on-screen timer counts the 10 minutes and warns at 1 minute left.

| Round | Time | Station 1 | Station 2 | Station 3 | Station 4 | Station 5 |
|---|---|---|---|---|---|---|
| **1** | 12:40 - 12:53 | Team 01 | Team 02 | Team 03 | Team 04 | Team 05 |
| **2** | 12:53 - 1:06 | Team 06 | Team 07 | Team 08 | Team 09 | Team 10 |
| **3** | 1:06 - 1:19 | Team 11 | Team 12 | Team 13 | Team 14 | Team 15 |
| **4** | 1:19 - 1:32 | Team 16 | Team 17 | Team 18 | Team 19 | Team 20 |
| **5** | 1:32 - 1:45 | Team 21 | Team 22 | Team 23 | Team 24 | Team 25 |

Suggested assignment: team numbers in order. Swap teams freely if a team needs more time to fix a problem, but keep a printed copy at every station so nobody waits.

**NOTE** This schedule needs **all 5 stations**. With only 4 stations, 25 teams take 7 rounds (about 91 minutes), which does not fit before 1:50. If a station fails, shorten play to 7 minutes per team or start demos earlier.

## One round at a station

| Phase | Station operator | Team |
|---|---|---|
| **Changeover**<br>3 min | Press R to release the previous team. Press T, check the next team is listed as online, type its number, press Enter. Seat the player about an arm's length away and press C to calibrate (look straight, eyes open, mouth closed). | Bring the laptop to the station. Listener running and showing CONNECTED. Game open and **clicked** so it has keyboard focus. |
| **Play**<br>10 min | Timer runs on screen. Press P to pause if the game crashes; press C again if the player changes. At 1:00 the timer turns red. | Team members and judges play using head gestures. |
| **End** | At 0:00 the screen shows TIME UP. Press R; keys are released at once. | Close the game and make room for the next team. |

## Gestures (for the 12:00 announcement and the stations)

| Gesture | Key |
|---|---|
| Tilt head up / down | Up / Down |
| Turn head left / right | Left / Right |
| Tilt and turn together | Diagonals, e.g. Up + Left |
| Open mouth | Z |
| Close both eyes a little longer than a normal blink (about 0.2 s) | Space |
| Hold the gesture | The key stays held until you return to neutral |
| Leave the camera view | All keys are released |

## 12:00 twist: what to tell the teams

Announce that the games will be played with head gestures, show one station live, then give teams these tips for the last 30 minutes:

- **Inputs are held, not tapped.** A key goes down when a gesture starts and up when it ends. Movement should respond to a held key, not need fast repeated presses.
- **Inputs are imprecise.** Players overshoot and react slower than with a keyboard. Slow the game down, make targets bigger and give more time to react.
- **Space is a deliberate blink.** It can't be pressed rapidly, so don't make Space something players must spam.
- **Z is an open mouth**, and holding the mouth open holds Z.
- **Diagonals are possible** (tilt and turn together), but single directions are easier to control.
- **Keys go to the focused window.** At the demo the game window must be clicked and in front.
- Test now: a volunteer will send test keys to your laptop before 12:30.

# Roles

| Role | People | Responsibilities |
|---|---|---|
| Tech lead | 1 | Relay server awake and healthy (open /status at about 7:50 AM). Runs the fallback relay if the internet fails. |
| Check-in desk | 2-3 | Help teams install the listener, run the connection test (listener --test shows CONNECTED), record each team's number. |
| Station operators | 5 (one per station) | Set up and calibrate stations by 12:30, run each demo round, keep time. |
| 12:00-12:30 key testers | 2-3 (desk volunteers) | Send test keys to each team's laptop and confirm they reach the game. |
| Judges | 2 faculty | Play the demos and choose the 2 winning teams. |

# Checklists

## Before the event

| | Item |
|---|---|
| ☐ | Relay server deployed and tested end to end (Testing Guide, Tests 0-6) |
| ☐ | 5 station laptops with webcams: station.py installed, face model downloaded, test calibration done |
| ☐ | Event token chosen and ready to hand out at check-in (not in the public repo) |
| ☐ | Listener instructions (PARTICIPANTS.md) ready for teams: printed or shared as a link |
| ☐ | Fallback: hotspot or router, and one laptop that can run relay.py (Testing Guide, Test 7) |
| ☐ | Printed demo schedule for every station; team numbers 01-25 assigned |
| ☐ | Power strips and extension cords for 5 stations and 25 team laptops |
| ☐ | Starter prompt finalized and ready to share at 9:00 |

## Check-in (8:00 - 9:00)

1. Give the team its number (01-25), the server address and the event token.
2. The team installs the listener (PARTICIPANTS.md, steps 1-3).
3. The team runs `python listener.py --team XX ... --test`. Pass = it shows `CONNECTED (team XX)`.
4. Mac users: make sure the terminal is allowed under Privacy & Security > Accessibility, or keys won't reach the game later.
5. Tick the team off on the check-in sheet.

> **NOTE** At check-in teams only see that a script will "send key presses to your game" during the demo. Don't explain how the keys are produced until the 12:00 reveal.

## 12:00 - 12:30 key test (each team)

1. The team runs the listener **without** `--test` and clicks its game window.
2. A volunteer sends test keys: `python tools/fake_station.py --server ... --token ... --team XX`.
3. Pass = the game reacts to arrows, Z and Space, and nothing stays held afterwards.

## Demo time (12:30 onward)

| | Item |
|---|---|
| ☐ | Open the relay /status page: all teams due to demo are listed as online |
| ☐ | Each station calibrated, named station1 to station5, and showing 25+ fps |
| ☐ | Schedule posted at each station; teams know their round and station |
| ☐ | Judges know which stations they will play at in each round |

Related documents: **Testing-Guide.pdf** (how to test the system), **ORGANIZERS.md** (station details and tuning), **PARTICIPANTS.md** (listener setup for teams), **DEPLOY.md** (relay server and fallback).